

Lab07_Mikhael

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SQLite - Parte II

1. Baixe o arquivo uwmadison.sqlite3. Conecte-se a ele usando o pacote RSQLite, armazenando a conexão em uma variável conn.

```
library(RSQLite)
conn <- dbConnect(SQLite(), "database.sqlite3")
```

2. Quem são e quantos são os professores que lecionaram disciplinas cujo tópico era estatística (subjects.abbreviation='STAT');

```
dbGetQuery(conn, "
SELECT COUNT(DISTINCT instructors.id) AS PROFESSORES
FROM subjects
INNER JOIN subject_memberships
  ON subjects.code = subject_memberships.subject_code
INNER JOIN course_offerings
  ON subject_memberships.course_offering_uuid = course_offerings.uuid
INNER JOIN sections
  ON course_offerings.uuid = sections.course_offering_uuid
INNER JOIN teachings
  ON sections.uuid = teachings.section_uuid
INNER JOIN instructors
  ON teachings.instructor_id = instructors.id
WHERE subjects.abbreviation = 'STAT'
")
```

```
PROFESSORES
1          379
```

3. O GPA americano é definido numa escala de 0 a 4, em que A = 4, AB = 3.5, B = 3, BC = 2.5, C = 2, D = 1 e F = 0. Determinando a nota média de cada oferecimento pela ponderação da quantidade de alunos em cada extrato com os valores numéricos de cada conceito, indique (no que se referente a disciplinas no assunto de estatística):

- Quem é o professor mais difícil?
- Quem é o professor mais fácil?
- Qual é a disciplina mais difícil?
- Qual é a disciplina mais fácil?

```
gpa_stat <- dbGetQuery(conn, "

WITH gpa AS (
  SELECT
    course_offering_uuid,
    section_number,
    (
      4.0 * a_count +
      3.5 * ab_count +
      3.0 * b_count +
      2.5 * bc_count +
      2.0 * c_count +
      1.0 * d_count +
      0.0 * f_count
    ) /
    (
      a_count +
      ab_count +
      b_count +
      bc_count +
      c_count +
      d_count +
      f_count
    ) AS gpa
  FROM grade_distributions
)

SELECT
  subjects.abbreviation,
  courses.number,
  courses.name AS disciplina,
  instructors.name AS professor,
  gpa.gpa
```

```

FROM subjects
INNER JOIN subject_memberships
  ON subjects.code = subject_memberships.subject_code
INNER JOIN course_offerings
  ON subject_memberships.course_offering_uuid =
    course_offerings.uuid
INNER JOIN courses
  ON course_offerings.course_uuid = courses.uuid
INNER JOIN sections
  ON course_offerings.uuid = sections.course_offering_uuid
INNER JOIN teachings
  ON sections.uuid = teachings.section_uuid
INNER JOIN instructors
  ON teachings.instructor_id = instructors.id
INNER JOIN gpa
  ON sections.course_offering_uuid = gpa.course_offering_uuid
  AND sections.number = gpa.section_number
WHERE subjects.abbreviation = 'STAT'
")

```

```

professores_stat <- dbGetQuery(conn, "
WITH gpa AS (
  SELECT
    course_offering_uuid,
    section_number,
    (
      4.0 * a_count +
      3.5 * ab_count +
      3.0 * b_count +
      2.5 * bc_count +
      2.0 * c_count +
      1.0 * d_count +
      0.0 * f_count
    ) /
    (
      a_count +
      ab_count +
      b_count +
      bc_count +
      c_count +
      d_count +
      f_count
    )

```

```

    ) AS gpa
FROM grade_distributions
WHERE
    a_count + ab_count + b_count + bc_count +
    c_count + d_count + f_count > 0
),
dados_stat AS (
    SELECT
        instructors.id AS professor_id,
        instructors.name AS professor,
        courses.number,
        courses.name AS disciplina,
        gpa.gpa
    FROM subjects
    INNER JOIN subject_memberships
        ON subjects.code = subject_memberships.subject_code
    INNER JOIN course_offerings
        ON subject_memberships.course_offering_uuid =
            course_offerings.uuid
    INNER JOIN courses
        ON course_offerings.course_uuid = courses.uuid
    INNER JOIN sections
        ON course_offerings.uuid = sections.course_offering_uuid
    INNER JOIN teachings
        ON sections.uuid = teachings.section_uuid
    INNER JOIN instructors
        ON teachings.instructor_id = instructors.id
    INNER JOIN gpa
        ON sections.course_offering_uuid = gpa.course_offering_uuid
        AND sections.number = gpa.section_number
    WHERE subjects.abbreviation = 'STAT'
)
SELECT
    professor,
    AVG(gpa) AS gpa_medio
FROM dados_stat
GROUP BY professor_id, professor
ORDER BY gpa_medio
")

#prof menor gpa
professores_stat[1, ]

```

```

        professor gpa_medio
1 JAMES D KUELBS      2.6375

```

```

#prof maior gpa
professores_stat[nrow(professores_stat), ]

```

```

        professor gpa_medio
207 YAJUAN SI         4

```

```

disciplinas_stat <- dbGetQuery(conn, "
WITH gpa AS (
  SELECT
    course_offering_uuid,
    section_number,
    (
      4.0 * a_count +
      3.5 * ab_count +
      3.0 * b_count +
      2.5 * bc_count +
      2.0 * c_count +
      1.0 * d_count +
      0.0 * f_count
    ) /
    (
      a_count +
      ab_count +
      b_count +
      bc_count +
      c_count +
      d_count +
      f_count
    ) AS gpa
  FROM grade_distributions
  WHERE
    a_count + ab_count + b_count + bc_count +
    c_count + d_count + f_count > 0
),
dados_stat AS (
  SELECT
    courses.number,
    courses.name AS disciplina,
    gpa.gpa

```

```

FROM subjects
INNER JOIN subject_memberships
  ON subjects.code = subject_memberships.subject_code
INNER JOIN course_offerings
  ON subject_memberships.course_offering_uuid =
    course_offerings.uuid
INNER JOIN courses
  ON course_offerings.course_uuid = courses.uuid
INNER JOIN sections
  ON course_offerings.uuid = sections.course_offering_uuid
INNER JOIN gpa
  ON sections.course_offering_uuid = gpa.course_offering_uuid
  AND sections.number = gpa.section_number
WHERE subjects.abbreviation = 'STAT'
)
SELECT
  number,
  disciplina,
  AVG(gpa) AS gpa_medio
FROM dados_stat
GROUP BY number, disciplina
ORDER BY gpa_medio
")

#disciplina menor gpa
disciplinas_stat[1, ]

```

```

      number                disciplina gpa_medio
1    431 Introduction to the Theory of Probability  2.891568

```

```

#disciplina maior gpa
disciplinas_stat[nrow(disciplinas_stat), ]

```

```

      number                disciplina gpa_medio
70    841 Nonparametric Statistics and Machine Learning Methods      4

```

4. Desconecte do banco de dados.

```
dbDisconnect(conn)
```

```
Sys.setenv(TZ = "America/Sao_Paulo")  
Sys.time()
```

```
[1] "2026-09-22 09:25:51 -03"
```